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AI-Based Real-Time Virtual Cloth Try-On System

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***Abstract*— The high growth rate in the development of electronic commerce in the fashion industry has had a significant impact on the way consumers purchase clothing. However, the absence of the opportunity to physically try the clothing before purchasing has led to inappropriate sizes and improper fit. In order to solve the problem of improper sizes and improper fit of clothing items, this paper proposes an AI-based Virtual Cloth Try-On system that enables the user to virtually try the clothing items on their own bodies. The proposed system utilizes a webcam or an image provided by the user to enable the system to be visible to the computer. Computer vision has been implemented in the proposed system. Computer vision enables the proper alignment of clothing items selected by the user and the alignment of clothing items on the user's body. It can be seen that the proposed system has shown smooth interaction with the users through the implementation of the proposed system. It can be seen that the proposed system has successfully aligned the clothing items selected by the user. It can be seen that the proposed system is a proper solution to solving the problem of improper sizes and improper fit of clothing items in the field of fashion technology at a low cost. It can be seen that the proposed system has shown the proper application of artificial intelligence in the field of fashion technology.**

Keywords: Virtual Try-On, Deep Learning, Image Processing, Pose Estimation, Computer Vision, E-commerce, AI

I. INTRODUCTION

The advent of online shopping sites has changed the functioning of the fashion industry. These online shopping sites offer users a feeling of convenience and variety. But the major problem that has arisen with the advent of online shopping sites is that the users are not getting a chance to virtually try the clothes. This has become a problem since the customers are not satisfied with the size and look of the clothes.

The advent of Virtual Try-on technology has become a major solution to the problem that has arisen in the fashion industry due to online shopping sites. This technology has allowed the users to virtually try the clothes. This technology has used computer vision and AI algorithms to virtually put the clothes on the body of the users. But the technology has also used complex devices such as augmented reality devices and 3D body scanners. This has become a problem since the technology cannot be used by the common public.

In this work, an AI-based Virtual Cloth Try-On system has been proposed. This system has used a standard webcam/image input device. This system has allowed the clothes to be virtually put on the body of the users by using pose estimation algorithms.

II. LITERATURE SURVEY

Virtual try-on technology has witnessed rapid development over the past few years. This development is attributed to both academia and industry. Initially, researchers were working on image-based approaches that tried to fit the garment over a person's image. One of the pioneering research papers in this field is VITON by Han et al. [1], which proposed a coarse-to-fine framework for generating clothing images over a person's body. The method is considered a milestone in the field of virtual try-on technology because it captured texture information effectively. Later, Wang et al. [2] proposed a new method called CP-VTON, which improved the garment fitting over a person's body by proposing a geometric transformation. The method also maintained important clothing information such as patterns and logos. However, these methods were not very effective for handling poses. The methods were primarily developed for frontal and static images only.

And Dong et al. [3] proposed a multi-stage virtual try-on method for handling multiple garments and poses. Around the same time, Güler et al. [4] proposed a significant method called DensePose, which maps a 2D image onto a 3D human body surface. The method provided a better understanding of the body structure, which is essential for a better fit.

Further improvements in the quality of output were achieved by the VITON-HD method by Choi et al. [5], which focused on achieving high-resolution images that are visually appealing. Ge et al. [6] introduced a parser-free method that removed the need for human parsing. Yang et al. [7] introduced the adaptive image generation method, which focused on preserving specific parts of the image. Further improvements in the system were made by Lee et al. [8] by separating the cloth warping and image generation. Context-aware methods such as C-VTON [9] have further enhanced the quality of images by considering the background. Recent advancements in the system have been made by the use of diffusion-based models such as CSD-VTON [10] and StyleVTON [11], which have further enhanced the quality of images by using generative techniques. Further improvements in the system have been made by the introduction of lightweight models for real-time applications [12]. Semantic guidance has further enhanced the quality of images by the introduction of FashionOn [13] and 3D models [14]. Further improvements in the system have been made by the introduction of lightweight models for real-time applications [15].

Despite the advancements in the system, challenges remain in the use of the system. The challenges in the system have been addressed by the introduction of a lightweight, real-time-based system for virtual try-on.

Research Objectives

1. To create a system that helps users try on clothes virtually with the help of the webcam or the image input.
2. To improve user experience by providing interactive and visually accurate results.
3. To create an effective system that helps the user try on clothes with the help of the user's body structure.
4. To apply computer vision and image processing techniques for realistic overlay of clothing on the user's image.
5. To create a cost-effective solution that can be easily used in online shopping platforms.

Problem Statement

Due to the rapid growth of online shopping for fashion wear, the customer is increasingly preferring to shop online. However, one major drawback of shopping online is the inability to try the clothing before purchasing. This, in turn, creates several problems for the customer. Due to this drawback, the customer loses trust in shopping online. Moreover, the chances of returns also increase. Although several solutions for the try-on option have been introduced in the field of online shopping, several drawbacks still exist. Most of the solutions require the purchase of high-end equipment, which is very costly. Moreover, the results obtained using these solutions are not accurate. Most of the solutions do not consider the pose, movement, and lighting conditions. Therefore, it is required to design and develop an efficient, cost-effective, and user-friendly try-on option for the customer using the techniques of artificial intelligence. It is required to design a system that can efficiently align the clothing with the individual's body using the webcam.

Scope of the study

The aim of this study is to develop and evaluate the efficacy of the proposed AI-based real-time virtual cloth try-on system, which can improve the online shopping experience by enabling the user to try on clothes virtually. It is meant for general users, considering the webcam or static images as input sources, which can reduce the cost of the required hardware. This research is centered on the application of computer vision and AI techniques to the simulation of clothing try-on. It takes into account the important factors related to the posture, movement, and lighting conditions of the human body. It is meant for the simulation of the try-on of clothing items such as tops, which is the main focus. It is related to the application

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of tools such as OpenCV and pose detection libraries to perform geometric transformations. It is evaluated on the basis of the alignment accuracy, real-time performance, and overall user experience.

The findings of this current research can be useful to the e-commerce sector to enhance the shopping experience and satisfaction level. It is centered on the application of computer vision techniques to the simulation of the try-on of clothing items. It is related to the application of tools such as OpenCV and pose detection libraries to perform geometric transformations. It is evaluated on the basis of the alignment accuracy, real-time performance, and overall user experience. It is limited to the simulation of the try-on of clothing items such as tops. It is related to the application of tools such as OpenCV and pose detection libraries to perform geometric transformations. It is evaluated on the basis of the alignment accuracy, real-time performance, and overall user experience.

III. METHODOLOGY

In this research work, a system-oriented and experimental methodology has been adopted to design and develop an AI-based real-time Virtual Cloth Try-On system. The methodology presented in this work incorporates computer vision techniques along with deep learning-based pose estimation models for simulating the visualization of a cloth on a human body.The input data required by this system is captured through a webcam or a static image provided by the user. Unlike other traditional machine learning-based systems, this methodology deals with real-time images, which are dynamic and interactive in nature. The system captures images in real-time, which are further processed using image processing techniques to extract meaningful features required for cloth fitting. The methodology presented in this work is mainly focused on detecting human body landmarks using pose estimation models. The landmarks represent significant points on a human body, such as shoulders, arms, torso, etc., which are required for understanding the posture and structure of a human body. Based on these detected points, spatial relationships are computed to determine the dimensions of a human body.

The workflow of the system involves several steps. First, the input is preprocessed to ensure the quality of the images is improved. It is followed.

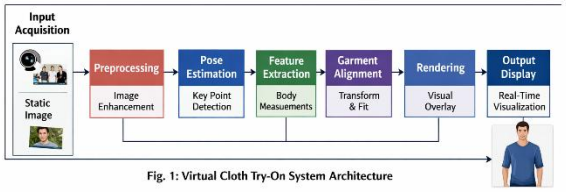

Fig.1 System Architecture

The system architecture shown in Fig.1 illustrates the complete workflow of the proposed Virtual Try-On system. It includes modules such as input acquisition, preprocessing, pose estimation, feature extraction, garment alignment, rendering, and output display. Each module interacts sequentially to transform raw input into a realistic virtual try-on output. The architecture highlights how computer vision techniques are integrated to achieve real-time performance and accurate garment fitting.

A. Pose Estimation Model Approach

In order to accurately align garments with the human body, the proposed system utilizes a pose estimation model to detect and map key body landmarks. Pose estimation plays a crucial role in understanding the structure, orientation, and movement of the human body from an input image or video frame. The system employs a pre-trained pose detection framework (such as MediaPipe Pose), which identifies a set of key points on the human body. These key points include major joints and reference locations such as shoulders, elbows, hips, and torso. Each landmark is represented by spatial coordinates in the 2D image plane.

The detected landmarks enable the system to track body posture dynamically and provide a structural representation of the user's body. This representation

is essential for determining how the garment should be scaled, positioned, and oriented.

Mathematically, the set of detected landmarks can be represented as:

$$L = \{l_1, l_2, l_3, \dots, l_n\}$$

Where:
$l_i = (x_i, y_i)$ represents the 2D coordinates of the ith body landmark n = total number of detected landmarks.

These landmarks act as anchor points for further processing and are used to compute body dimensions and spatial relationships.

B. Feature Extraction and Body Modeling

Once the body landmarks are detected, the system performs feature extraction to derive meaningful measurements required for garment fitting. These features include body width, height, orientation, and relative distances between key landmarks.

For example, the width of the upper body can be estimated using the distance between shoulder points:

$$W = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Similarly, the height and orientation of the body are calculated using vertical and angular relationships between landmarks.

These extracted features form a simplified body model, which is used as a reference for aligning the garment. The accuracy of these features directly impacts the realism of the virtual try-on output.

C. Garment Alignment Model

The garment alignment model is centered on transforming the selected clothing image to ensure it fits naturally and accurately on the user's body. This is achieved through the application of geometric transformation techniques based on the derived body features, including the body width, height, and orientation. The alignment is done to ensure the garment is transformed dynamically to fit the user's posture, which enhances the realism of the virtual try-on experience. The transformation process involves three main steps. Scaling is applied to transform the size of the garment to fit the user's body size. This is done to ensure the garment is not stretched when it is transformed to fit the user's body. The translation process is applied to ensure the garment is translated to the right position on the body. For instance, the garment is translated to the shoulders and torso region. Finally, the rotation process is applied to ensure the orientation of the garment is similar to the orientation of the body.

Through the application of these processes, the garment is effectively aligned with the user's body structure.

Mathematically represented as

$$I' = T(I)$$

$I' \rightarrow$ Original garment image

$T \rightarrow$ Transformation function

$I' \rightarrow$ Transformed (aligned) garment image

The transformation function T includes scaling factor (s), rotation angle (θ), and translation vector (t).

D. Image Rendering and Overlay

After aligning the garment, the system overlays it onto the user's image using image processing techniques. This step ensures that the final output appears realistic and visually appealing. The rendering process involves Alpha blending which is combining garment and background images , Masking where it removing unwanted regions of the garment image and Edge Smoothing reducing sharp boundaries for natural appearance

Mathematically represented as:

$$O = \alpha G + (1 - \alpha)B$$

$O \rightarrow$ *Final output image*

G → Garment image (shirt, dress, etc.)

B → Background image (user's body/frame)

α → Transparency factor (ranges from 0 to 1)

This step enhances the realism of the virtual try-on system.

E. Working of the Proposed System

The proposed Virtual Cloth Try-On system is based on a sequential process to produce output in real time. Initially, the input is captured by the webcam or the image and processed frame by frame. Each frame is processed by the pose estimation module to identify key body landmarks. Once the landmarks are detected for the body, the spatial relationships between different body parts are calculated. The calculated features are then used to calculate the correct size, position, and orientation of the selected garment.

The garment image is processed by performing different operations such as scaling, translating, and rotating the garment to align it according to the body structure of the user. Once the garment is processed and aligned according to the body structure of the user, it is blended over the user's image to produce a realistic output. The process is continuously repeated for every frame of the video captured by the webcam. The position of the garment is dynamically updated based on the movement of the user to produce a realistic output.

F. Data Processing and Feature Computation

In the proposed system, data processing is performed on visual input rather than traditional datasets. The captured frames undergo preprocessing steps such as resizing, noise reduction, and color normalization to improve detection accuracy. The primary data used in the system consists of body landmark coordinates obtained from the pose estimation model. These coordinates are processed to compute essential features like Distance between key landmarks (e.g., shoulder width) and Body orientation and angle .

These computed features are used to guide the garment alignment process. Unlike traditional scoring methods, the system dynamically calculates these values in real time for each frame. The feature computation ensures that the garment adapts accurately to different users, body shapes, and movements.

G. System Implementation and Model Integration

The proposed system is implemented using Python and integrates multiple computer vision and AI-based libraries. OpenCV is used for image processing and real-time video handling, while MediaPipe is used for pose estimation and landmark detection.

The system is designed as a real-time application that continuously processes video frames and updates the output dynamically. The implementation includes the following components:

- Input module for webcam/image capture
- Pose detection module for extracting body landmarks
- Feature computation module for calculating body dimensions
- Garment alignment module using geometric transformations
- Rendering module for overlaying garments
- Output display module for real-time visualization

The system does not rely on heavy deep learning training but instead uses pre-trained models for efficient performance. This makes the system lightweight and suitable for execution on standard hardware without requiring GPUs.

H. Performance Evaluation

The performance of the proposed system is evaluated using multiple criteria to ensure its effectiveness, accuracy, and usability in real-time applications. Alignment accuracy measures how precisely the garment fits onto the user's body based on the detected landmarks, ensuring proper scaling and positioning. Real-time processing evaluates the system's ability to handle continuous input frames efficiently without noticeable delay, which is crucial for providing a smooth and interactive user experience. Visual realism

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assesses the quality of the garment overlay by examining factors such as edge smoothness, blending quality, and the natural appearance of the clothing on the user's body. Additionally, robustness analyzes the system's performance under varying conditions, including changes in lighting, dynamic body movements, and partial occlusions. Experimental results indicate that the system performs efficiently in real-time scenarios and delivers a satisfactory user experience. However, minor inaccuracies may occur during rapid movements or under poor lighting conditions, which can slightly affect alignment and rendering quality.

IV. Results and Discussions

In this paper, the implementation and evaluation of the proposed AI-based Real-Time Virtual Cloth Try-On using the Virtual Try-On system are discussed. The proposed system is evaluated using different images for the garment and the model.

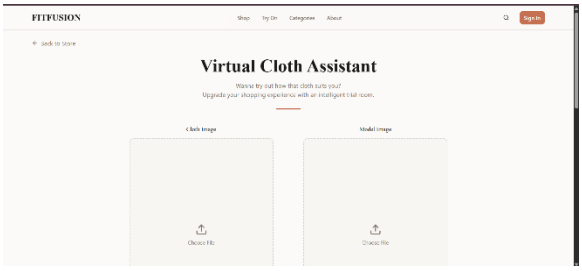

Fig.2 User Interface and Input Module

Figure 2 shows the initial interface for the proposed Virtual Try-On system. In the interface, the user is allowed to input the images for the garment and the model.

In Figure 2, the input module for the proposed Virtual Try-On is shown. During this module, the user is allowed to input the images for the garment and the model. The images can be input into the system using the placeholders for the "Cloth Image" and the "Model Image."

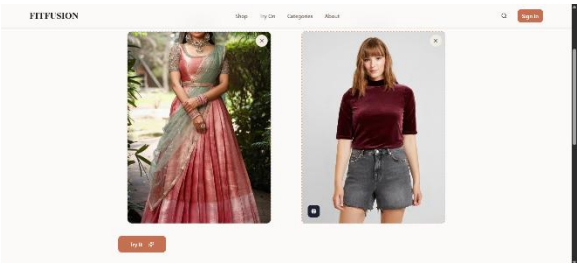

Fig.3 Input Processing and Try-On Initialization

Once the input images have been chosen, the system displays these images, as shown in Fig. 3. This phase involves the preparation of the garment image and the model image. The user can then start the virtual try-on process by clicking the "Try It" button.

Fig. 3 illustrates the intermediate phase of the system, in which the two images are visible. The system internally undergoes a preprocessing phase, pose detection, and feature extraction during this phase. This phase is important because the images are being prepared for alignment.

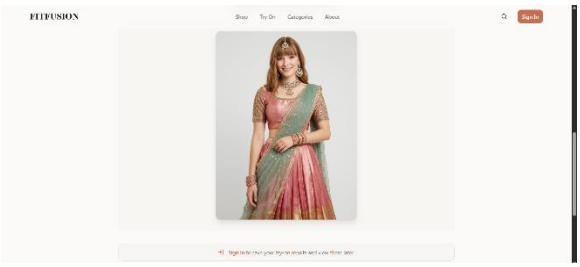

Fig.4 Final Output

The final output of the system can be observed in Fig. 4, in which the chosen garment has been successfully integrated with the model image. The system aligns the garments with the body of the user by considering the body landmarks.

Fig. 4 illustrates the output of the proposed system, in which the garments have been successfully integrated with the model image. The proposed system has been successful in producing an output that resembles a

real-world scenario. The output has been aligned perfectly, maintaining a natural appearance. The blending techniques used in the proposed system have resulted in smooth edges, which have reduced the visual artifacts.

V. CONCLUSION

In this paper, the design and implementation of an AI-based Real-time Virtual Cloth Try-On system have been presented. The system allows users to virtually try on clothing on their bodies. The system has been designed by applying computer vision techniques. The proposed system has been designed in such a way that it can utilize pose estimation, image processing, and geometric transformations for accurate clothing alignment on the human body. The complete framework has been developed for the system, which includes acquisition, preprocessing, pose detection, feature extraction, garment alignment, and rendering. The major contribution of this research is that the system can provide a realistic and interactive Virtual Try-On facility for users by utilizing a webcam/image input. The system can successfully detect body landmarks for garment alignment. The results have shown that the proposed system can effectively simulate clothing fit on the human body. The system does not require expensive hardware such as 3D scanners or augmented reality devices for clothing alignment. The experimental results have shown that the system can perform well in terms of accuracy, response time, and visual quality. The pose estimation has been integrated into the system for dynamic garment adaptation according to user movements. The rendering techniques have been used in the system for producing visually appealing images with minimal distortion. The major contribution of this research is the practical use of artificial intelligence in the domain of Fashion Technology. The system can provide a cost-effective solution for the e-commerce sector. The system can help users in reducing uncertainty during online shopping. The system can provide a Virtual Try-On facility for users, which can help in reducing return rates. The system can increase customer satisfaction.

However, the system also has some limitations. The current system is based on 2D image processing. The accuracy of the alignment may also be affected if the conditions are such that the body is in an extreme pose or is moving very fast. The system also performs better if the garments are simple. The accuracy of the alignment may also be poor if the garment is complex or if it is a combination of multiple garments.

Scope for Further Research

There are several scopes for further research in this area. The accuracy of the system may be improved by implementing some of the latest developments in deep learning and 3D modeling techniques. The integration of 3D body and garment simulation may also improve the accuracy of the output.

Further research may also be carried out by implementing high-resolution image synthesis models and diffusion-based approaches for better results. The system may also be extended to include different types of garments such as loose garments and garments that are a combination of multiple garments.

Further research may also be carried out by developing mobile-based applications and cloud integration for better results. The integration of real-time user feedback and personalization may also improve the overall user experience.

The research paper highlights the potential of artificial intelligence and computer vision for revolutionizing the online shopping experience. The integration of virtual try-on systems may become an essential part of modern online shopping systems.

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